

Your grade: 100%

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Next item →

1. What communication model is MPI most fundamentally built on?

1 / 1 point

- ☐ Centralized communication - messages are passed to one particular processor which distributes the messages to their destination processors
- ☐ Message pooling - processors send data to a "message pool" from which messages are retrieved regularly by each processor on the network
- ☒ Pairwise communication - messages are passed directly from a sending processor to another processor which explicitly receives the message
- ☐ Broadcast communication - messages are sent to all processors on the network, with a tag indicating which processor the message is meant for; only this destination processor will receive and act on the message

☒ Nice work

Correct: The first MPI standard (v1) in 1994 introduced basic point to point message passing.

2. Which of the following may be factors that hamper the speedup a program run on multiple cores?

1 / 1 point

- ☐ Communication overhead between cores
- ☐ Indivisibility of work
- ☐ Memory bottlenecks
- ☒ All of the above

☒ Nice work

Correct: All are important factors.

3. When using a naïve point-to-point communication strategy to broadcast data from each rank to each other rank, how would the time required scale with the data?

1 / 1 point

- ☐ Logarithmically
- ☐ Linearly
- ☒ Quadratically
- ☐ Remains constant

☒ Nice work

Correct: Each of the N ranks handshake with N-1 neighbors.

4. What is the primary advantage of using nonblocking communications in MPI?

1 / 1 point

- ☐ It allows for faster data transmission
- ☒ It enables the program to proceed with computations independent of communication buffers
- ☐ It eliminates the need for ghost zones
- ☐ It automatically balances the workload across all processes

☒ Nice work

Correct: this allows for overlapping communication and computation, which is the main advantage of nonblocking communication.

5. In the context of nonblocking MPI communication, what does the term "asynchronous transmission" mean?

1 / 1 point

- ☐ Data is transferred only during MPI_wait calls
- ☒ Data is transferred independent of additional MPI library calls
- ☐ Data is transferred only between adjacent processes
- ☐ Data is transferred only at the end of the program

☒ Nice work

Correct: this is the definition of asynchronous transmission in MPI.

6. Which function is used to verify the completion of a single nonblocking message in MPI?

1 / 1 point

- ☐ MPI_send
- ☐ MPI_recv
- ☒ MPI_wait
- ☐ MPI_waitall

☒ Nice work

Correct: MPI_wait is used to verify the completion of a single nonblocking message.

7. What is a potential issue when using MPI_waitall for nonblocking communications?

1 / 1 point

- ☐ It always causes a deadlock
- ☒ It can result in a deadlock if called by processes not participating in communication
- ☐ It slows down the program significantly
- ☐ It cancels all ongoing communications

☒ Nice work

Correct: this is a pernicious deadlock scenario mentioned in the lecture.

8. In the context of load balancing in MPI, what are the three main categories mentioned?

1 / 1 point

- ☒ Task-based, geometry-based, and mixed
- ☐ Static, dynamic, and adaptive
- ☐ Local, global, and hybrid
- ☐ Synchronous, asynchronous, and semi-synchronous

☒ Nice work

Correct: these are the three categories of load balancing discussed in the lecture.

9. What is a common example of task-based load balancing in MPI?

1 / 1 point

- ☐ Dividing a matrix evenly among all processes
- ☒ Using some ranks for computation and others for I/O to storage
- ☐ Assigning each process a single column of data
- ☐ Dynamically redistributing work every iteration

☒ Nice work

Correct: this is a common task-based load balancing strategy mentioned in the lecture.

10. In geometry-based load balancing for a 2D problem, what is a potential drawback of dividing the domain into too many small subdomains?

1 / 1 point

- ☐ Increased memory usage
- ☐ Reduced accuracy of calculations
- ☒ Increased communication overhead
- ☐ Slower individual process speed

☒ Nice work

Correct: dividing into too many small subdomains can lead to communication overhead outweighing computational benefits.

11. How does the computation time typically scale with the number of MPI ranks in an ideal case?

1 / 1 point

- ☐ Linearly
- ☐ Exponentially
- ☐ Logarithmically
- ☒ Inversely

☒ Nice work

Correct: in an ideal case, computation time is inversely proportional to the number of MPI ranks.

12. What is the recommended order of operations when implementing overlapping communication and computation in MPI?

1 / 1 point

- ☐ Isends, trevs, interior work, wait, boundary work
- ☒ trevs, isends, interior work, wait, boundary work
- ☐ Wait, isends, trevs, interior work, boundary work
- ☐ Interior work, isends, trevs, wait, boundary work

☒ Nice work

Correct: this is the recommended order to maximize overlap and avoid potential issues.

13. How does the communication time typically scale in a 2D domain decomposition as the number of MPI ranks increases?

1 / 1 point

- ☐ It remains constant
- ☐ It increases linearly
- ☒ It decreases inversely proportional to the number of ranks
- ☐ It increases logarithmically

☒ Nice work

Correct: in 2D decomposition, communication components decrease inversely proportional to the number of ranks in each dimension.